

analytical summary

I used titanic data set and I find that it have tow main issues a lot of missing values on cabin and null values on age that I solve by drop the cabin because of that most of it have null values and fill null values of age by the mean of ages the lie chart was good but have problem that Pclass is float on the chart so I had to use xticks to make it integer

The distribution of passenger ages (histplot) shows that the majority of passengers were young adults between 20 and 40 years old, with a mean age of approximately 30 years

There are fewer elderly passengers above 60, suggesting that most travelers were in their prime working age. The histogram also highlighted a slight concentration of children under 10, indicating families traveling together. This distribution provides context for understanding survival trends relative to age ,The categorical analysis of survival by sex (countplot) illustrates a pronounced disparity between male and female survival rates. Female passengers were far more likely to survive than male passengers, confirming historical accounts of “women and children first” during the evacuation. The bar heights indicate that while male passengers were the majority, a larger proportion of females survived, suggesting gender played a critical role in survival probability

The scatterplot of Age versus Fare, colored by survival, highlights a correlation between fare paid and survival. Passengers who paid higher fares, often corresponding to first-class cabins, were more likely to survive. The plot also shows that younger passengers had a wide range of fares, but the highest survival rates were concentrated among those with expensive tickets, suggesting socio-economic status influenced access to safety during the disaster

Finally, the pairplot of Age, Fare, Pclass, and Survival provides a multi-dimensional view, reinforcing earlier insights. It shows that lower-class passengers paid less and had lower survival rates, while higher-class passengers with higher fares were more likely to survive. Age appears less strongly correlated with survival when controlling for class and fare, but younger passengers in higher classes still show better outcomes.

Data-driven insights and recommendations:

1. Socio-economic status, reflected by fare and class, was a strong determinant of survival; emergency planning should consider equitable access to safety resources.

2. Gender influenced survival outcomes; future protocols should account for vulnerable populations in evacuation strategies.
3. Young adults formed the majority of travelers; age-based risk assessments can help target safety interventions effectively.